Answers learning task 3

2026 – 2027

For exercises 3.1 and 3.5, determine the range (possible values) of the random variable.

Exercise 3.1: The random variable is the number of nonconforming solder connections on a printed circuit board with 2000 connections.

Answer

$\{0, 1, 2, \ldots, 2000\}$

Exercise 3.5: A batch of 600 machined parts contains 10 that do not conform to customer requirements. Parts are selected successively, without replacement, until a nonconforming part is obtained. The random variable is the number of parts selected.

Answer

$\{1, 2, 3, \ldots, 591\}$

Exercise 3.14: The sample space of a random experiment is $\{a, b, c, d, e, f\}$, and each outcome is equally likely. A random variable is defined as follows:

Outcome	a	b	c	d	e	f
x	0	0	1.5	1.5	2	3

Determine the probability mass function of X. Use the probability mass function to determine the following probabilities:

(a) $P(X = 2)$
(b) $P(0.6 < X < 2.7)$
(c) $P(X > 3)$
(d) $P(0 < X \leq 2)$
(e) $P(X = 0 \text{ or } X = 2)$

Answer

(a) 0.1667
(b) 0.5
(c) 0
(d) 0.8333
(e) 0.5

Exercise 3.17: Verify that the following function is a probability mass function, and determine the requested probabilities.

$$f(x) = \frac{2x + 1}{25}, \qquad x = 0, 1, 2, 3, 4$$

(a) $P(X = 3)$
(b) $P(X \leq 1)$
(c) $P(2 \leq X < 4)$
(d) $P(X > -10)$

Answer

This function is a probability mass function, since for each possible value of x the function value $f(x) \geq 0$. Furthermore, the sum for $x = 0, 1, 2, 3, 4$ of $f(x)$ equals 1.

(a) 0.28
(b) 0.16
(c) 0.48
(d) 1

Exercise 3.29: An assembly consists of three mechanical components. Suppose that the probabilities that the first, second, and third components meet specifications are 0.95, 0.98, and 0.97, respectively. Assume that the components are independent. Determine the probability mass function of the number of components in the assembly that meet specifications.

Answer

x	0	1	2	3
$f(x)$	0.00003	0.00301	0.09389	0.90307

Exercise 3.35: Determine the cumulative distribution function of the random variable with the following probability mass function:

x	-2	-1	0	1	2
$f(x)$	$1/8$	$2/8$	$2/8$	$2/8$	$1/8$

Furthermore, determine the following probabilities:

(a) $P(X \leq 1.25)$
(b) $P(X \leq 2.2)$
(c) $P(-1.1 < X \leq 1)$
(d) $P(X > 0)$

Answer

(a) 0.875
(b) 1
(c) 0.75
(d) 0.375

Exercise 3.43: Verify that the following function is a cumulative distribution function, and determine the probability mass function and the requested probabilities:

$$F(x) = \begin{cases} 0 & x < -10 \\ 0.3 & -10 \leq x < 30 \\ 0.7 & 30 \leq x < 50 \\ 1 & 50 \leq x \end{cases}$$

(a) $P(X \leq 50)$
(b) $P(X \leq 40)$
(c) $P(40 \leq X \leq 60)$
(d) $P(X < 0)$
(e) $P(0 \leq X < 10)$
(f) $P(-10 < X < 10)$

Answer

(a) 1
(b) 0.7
(c) 0.3
(d) 0.3
(e) 0
(f) 0

Exercise 3.51: Determine the mean and variance of the following random variable:

x	-2	-1	0	1	2
$f(x)$	$1/8$	$2/8$	$2/8$	$2/8$	$1/8$

Answer

$E(X) = 0, \quad V(X) = 1.5$

Exercise 3.53: Determine the mean and variance of the following random variable with the following probability mass function:

$$f(x) = \frac{2x+1}{25}, \qquad x = 0, 1, 2, 3, 4$$

Answer

$E(X) = 2.8, \quad V(X) = 1.36$

Exercise 3.67: Let the random variable X have a discrete uniform distribution on the integers $1 \leq x \leq 5$. Determine the mean and variance of X.

Answer

$E(X) = 3, \quad V(X) = 2$

Exercise 3.68: Thickness measurements of a coating process are made to the nearest hundredth of a millimeter. The thickness measurements are uniformly distributed with values 0.14, 0.15, 0.16, 0.17, 0.18, 0.19, and 0.20. Determine the mean and variance of the coating thickness for this process.

Answer

$E(X) = 0.17, \quad V(X) = 0.0004$

Exercise 3.70: The lengths of plate glass parts are measured to the nearest tenth of a millimeter. The lengths are uniformly distributed with values at every tenth of a millimeter starting at 640.0 and continuing through 640.9. Determine the mean and variance of the lengths.

Answer

$E(X) = 640.45, \quad V(X) = 0.0825$